MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 578

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Rectifier, half-wave, high vacuum

See figure 1

Mounting position: Any

Weight: 4.0-ounces (113.4 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	ib	Ib	tk	Pp
Unit:	Vac	kv	ma	mAdc	sec	W
Maximum:	5.25	40	750	100	---	75
Minimum:	4.75	---	---	---	5	---
TEST CONDITIONS:	5.0	---	---	---	5	---

GENERAL:

Qualification - Required

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1231	Peak emission	-	eb = 4,000 v (max)	0.65	II	is	1.6	---	a
1301	Filament current	-		0.65	II	If	5.50	6.50	Aac
1353	Operation of rectifiers (1)	1	epx = 40 kv; ib = 600 ma (min); Ib = 100 mAdc	0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1036	Bump	-	Angle = 25°	---	---	---	---	---	---
1256	Electrode current (anode)	-	Eb = 200 Vdc	---	---	Ib	75	---	mAdc
	Quality conformance inspection, part 3								
---	Life test	-	Group C; operation of rectifiers (1); t = 500 hours	---	---	---	---	---	---
---	Life-test end point:								
1231	Peak emission and failure to rectify	-		---	---	is	1.2	---	a
1353	Operation of rectifiers (2)	2	epx = 40 kv; ib = 750 ma; Ib = 100 mAdc	---	---	---	---	---	---

NOTES:

1. This test is to be the first test performed at the conclusion of the holding period.

(C) 2. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

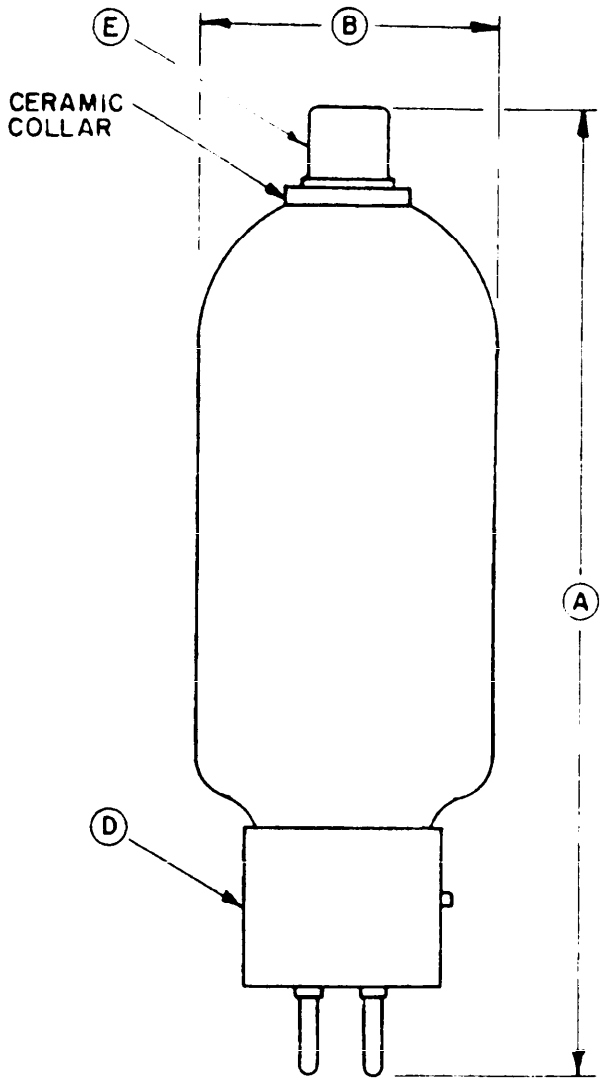
User activities:
Army - MU, WC
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2927)

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PIN CONNECTIONS	
Pin No.	Element
1	f
2	nc
3	nc
4	f
cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note)	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	6.000 (152.40)	6.500 (165.10)
B		2.070 (52.58)
C Quality conformance inspection, part 3 (see note)		
D	Base: A4-10 (EIA)	
E	Cap: C1-5 (EIA)	

NOTE:
These dimensions shall be checked annually with the following sampling plan:

n1 = 4 c1 = 0
n2 = 4 c2 = 1;

where c2 represents the total allowable failures for the first and second samples combined.

In case of failure after double sampling, the failing dimension(s) shall become quality conformance inspection (QCI), part 2, for three successful consecutive submissions, at which time the test may revert to the QCI, part 3, basis.

FIGURE 1. Outline drawing of electron tube type 578.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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OMB Approval
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